FINAL REPORT

AirSense Pakistan: AI-Powered Real-Time Air Quality Monitoring and Prediction System

Abstract

Air pollution has become one of the most serious environmental and public health challenges in Pakistan. Citizens often lack access to accurate, real-time information about air quality in their cities and neighborhoods. AirSense Pakistan is an AI-powered web application designed to monitor, visualize, and predict air quality conditions across major cities in Pakistan. The system integrates real-time environmental data, interactive maps, air quality forecasting, health risk assessment, and an AI chatbot assistant to help users understand pollution levels and make informed decisions regarding outdoor activities and health protection.

The application uses modern web technologies, machine learning concepts, mapping systems, and environmental APIs to provide a comprehensive air quality monitoring solution.

Chapter 1: Introduction

1.1 Background

Air pollution is a growing concern in Pakistan, especially in major cities such as Lahore, Karachi, Islamabad, Faisalabad, and Multan. Pollutants such as PM2.5, PM10, carbon monoxide, nitrogen dioxide, and ozone can negatively affect human health.

Many existing solutions provide only general air quality information and do not offer localized visualizations, health recommendations, or AI-powered assistance.

1.2 Problem Statement

Citizens lack a centralized platform that can:

- Monitor real-time air quality.
- Visualize pollution distribution across city areas.
- Predict future air quality conditions.
- Provide personalized health recommendations.
- Explain environmental information in simple language.

1.3 Project Objectives

The objectives of AirSense Pakistan are:

- Monitor real-time air quality data.

- Display pollution levels on an interactive map.
- Show area-wise pollution distribution.
- Predict future AQI trends.
- Provide personalized health risk assessments.
- Integrate an AI assistant for environmental awareness.
- Support multiple Pakistani cities.

Chapter 2: Literature Review

Existing Solutions

AirVisual

Provides global AQI information but lacks localized Pakistani area-level visualization.

OpenWeather Air Pollution API

Provides pollutant concentration data but does not offer health prediction systems.

WAQI (World Air Quality Index)

Offers AQI monitoring but limited personalized health guidance.

Research Gap

Existing platforms generally lack:

- Area-wise pollution visualization.
- AI-powered environmental assistant.
- Personalized risk analysis.
- Localized Pakistan-focused implementation.

AirSense Pakistan addresses these limitations.

Chapter 3: System Analysis

Functional Requirements

The system shall:

- Search Pakistani cities.
- Retrieve real-time environmental data.
- Display AQI values.
- Display PM2.5 and PM10 levels.
- Show temperature and humidity.

- Visualize pollution on maps.
- Predict AQI trends.
- Assess health risks.
- Provide AI chatbot assistance.

Non-Functional Requirements

- Fast response time.
- User-friendly interface.
- Responsive design.
- High availability.
- Data accuracy.
- Scalability.

Chapter 4: System Design

System Architecture

Frontend

- React.js
- Vite
- Tailwind CSS
- Leaflet Maps
- Recharts

Backend

- FastAPI
- Python

APIs

- OpenWeather API
- WAQI API
- Gemini AI API

Data Storage

- CSV Dataset
- GeoJSON Files
- Local Configuration Files

Use Case Diagram

User

- Search city

- View AQI
- View pollution map
- Check health risk
- View forecast
- Ask AI assistant

Chapter 5: Implementation

Search System

Users can search Pakistani cities through a dynamic search bar.

Supported cities include:

- Lahore
- Karachi
- Islamabad
- Faisalabad
- Multan
- Peshawar
- Quetta

Real-Time Monitoring

The system fetches:

- AQI
- PM2.5
- PM10
- Temperature
- Humidity
- Wind Speed

using environmental APIs.

Pollution Map

The pollution map is implemented using Leaflet.

Features include:

- Interactive city map.
- Area-wise AQI visualization.
- Color-coded pollution zones.

AQI Color Scheme

AQI Range	Color	Status
0-50	Green	Good
51-100	Yellow	Moderate
101-150	Orange	Unhealthy Sensitive
151-200	Red	Unhealthy
201+	Dark Red	Very Dangerous

AQI Forecast

The application predicts AQI conditions for:

- Current Time
- Next 6 Hours
- Next 24 Hours

The forecast is displayed using interactive charts.

Health Risk Assessment

Users can enter:

- Age
- Asthma Status
- COPD Status
- Smoking Status
- Heart Disease Status
- Outdoor Job Status

The system evaluates risk and generates recommendations.

AI Chatbot

The AI assistant is powered by Google Gemini.

Capabilities include:

- Explaining AQI.
- Explaining PM2.5.
- Health guidance.
- Environmental awareness.
- Pollution-related questions.

Chapter 6: Results

The developed system successfully provides:

Real-Time Dashboard

- AQI monitoring
- Weather monitoring
- Pollution indicators

Interactive Mapping

- City selection
- Area-wise pollution display
- AQI color visualization

Forecasting

- Future AQI predictions
- Trend visualization

Health Risk Analysis

- Personalized recommendations
- Risk classification

AI Assistance

- Interactive environmental chatbot

Chapter 7: Advantages

- Real-time environmental monitoring.
- User-friendly dashboard.
- AI-powered assistance.
- Health-focused recommendations.
- Interactive map visualization.
- Pakistan-focused implementation.
- Responsive design.

Chapter 8: Limitations

- Depends on external APIs.
- Some city areas use estimated AQI values.
- Prediction accuracy depends on available environmental data.
- Requires internet connectivity.

Chapter 9: Future Enhancements

Future versions may include:

- Satellite pollution monitoring.
- Machine learning AQI prediction models.
- Mobile application.
- Push notifications.
- Smart wearable integration.
- Government environmental reporting.
- Historical pollution analytics.
- District-level pollution mapping.

Conclusion

AirSense Pakistan successfully demonstrates how modern web technologies, environmental APIs, artificial intelligence, and geospatial visualization can be integrated to create a smart environmental monitoring platform. The system provides real-time air quality monitoring, pollution visualization, AQI forecasting, health risk assessment, and AI-powered assistance. The project contributes towards environmental awareness and helps citizens make informed health and lifestyle decisions based on current air quality conditions.